Transition of Care AI-Assisted Proof of Concept Leveraging the Clinical Health AI-augmented Records Toolkit

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PACIO Project

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Assembling structured and unstructured data into clinically relevant information using Generative AI, HL7® FHIR®, and the PACIO Transitions of Care IG.

* PACIO is the Post-Acute Care InterOperability Project.

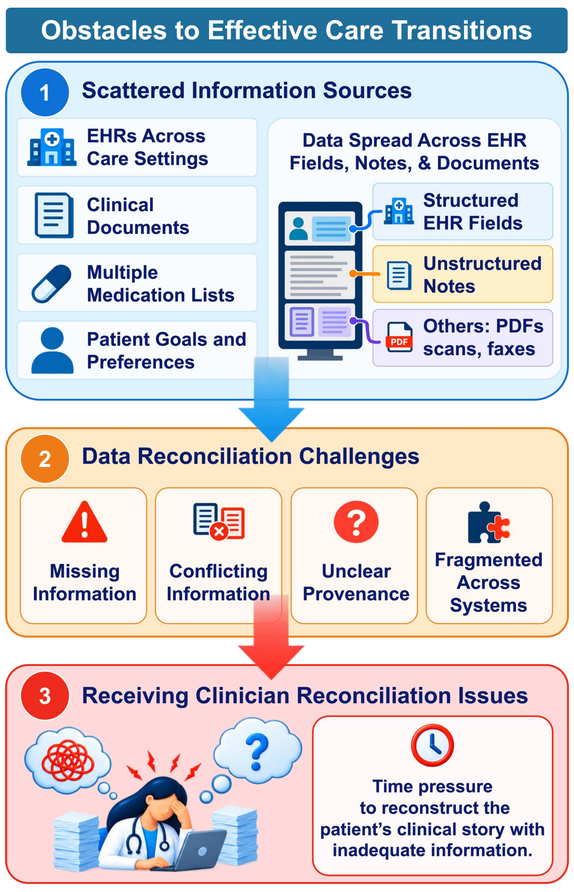
Introduction

Clinical problem

Post-acute transitions are often hindered by incomplete, fragmented, and inconsistently represented handoff information. Clinical data are scattered across EHR fields, notes, and documents, resulting in duplication, conflicting information, unclear data provenance, and missing context. Receiving teams must often reconstruct the patient's clinical story with ***inadequate information***.

Why standards alone are insufficient

PACIO and FHIR enable standardized data exchange but do not inherently support clinical interpretation. They do not filter for relevance, reconcile conflicting or duplicate data, assess completeness or accuracy, prioritize actionable information, or provide the person-centric view needed for effective care transitions.



Materials & Methods

Synthetic post-acute patient scenarios combining structured and free-text clinical records were used to develop and evaluate our prototype. Development incorporated input from the PACIO ToC subgroup and was guided by FHIR, USCDI v5, and the PIQI Framework. **Our prototype consisted of two complementary systems**:

- **CHART (C**linical **H**ealth **A**I-augmented **R**ecords **T**oolkit**)**, a reusable modular AI toolkit for clinical information extraction, analysis, synthesis, and transformation
- **ToC-AI (T**ransitions **o**f **C**are **AI-**Assisted**)** Application, a proof-of-concept implementation that used CHART capabilities to orchestrate end-to-end transitions of care.

AI-assisted data processing workflow

- Queried and ingested structured and unstructured EHR data.
- Extracted clinically relevant entities mapped to USCDI v5 elements.
- Applied element-level governance to support transition-level information generation.
- Normalized to standard terminologies as applicable.
- Detected duplicate and conflicting information.
- Presented flagged information for clinician review and reconciliation.
- Packaged reconciled content into a PACIO ToC FHIR IG-aligned document and bundle.

Clinician review and reconciliation

- Clinicians reviewed AI-suggested content.
- Assessed clinical relevance, source support, and appropriateness for inclusion.
- Resolved conflicting or inconsistent information.
- Determined readiness for downstream sharing.

Findings

Functional proof-of-concept findings

ToC-AI extracted structured and free-text data from synthetic scenarios, identified and mapped key USCDI v5 elements, detected duplicate and conflicting information, and produced quality-scored, clinician-reviewable transition data. The prototype also generated FHIR bundles aligned with the PACIO Transitions of Care (ToC) Implementation Guide (IG).

Clinical workflow relevance

The prototype consolidated fragmented source data into a unified handoff view, surfaced key transition information and potential risks, and preserved provenance for clinician review. This approach may reduce manual chart-review burden, although that outcome was not measured in the proof of concept.

Informatics insight

The exploration suggests the value of interoperability extends beyond data movement to semantic structuring, relevance filtering, and interpretability. By organizing standards-based data into clinically meaningful information, the approach bridges the gap between FHIR exchange and clinical usability.

Evidence boundary

Findings reflect synthetic proof-of-concept scenarios only. No real-world deployment, clinical outcomes, or efficiency measures were assessed, and potential reductions in chart-review burden remain hypotheses for future testing. The prototype demonstrated technical feasibility rather than clinical effectiveness.

Evidence Boundary: Proof of Concept vs. Future Exploration

Evaluation Area	✓ Demonstrated in Proof of Concept	◷ To be Measured in Future Exploration
Data inputs	✓ Processed structured and unstructured synthetic PAC data.	◷ Real-world performance across heterogeneous PAC data sources.
Clinical extraction & mapping	✓ Extracted relevant data and identified USCDI v5 / PACIO ToC elements.	◷ Accuracy, recall, and completeness against the PACIO ToC IG benchmark.
Duplicate / conflict reconciliation	✓ Identified duplicate and conflicting information in test scenarios.	◷ Conflict-detection precision and clinician agreement with AI suggestions.
Confidence & source traceability	✓ Assigned confidence scores and preserved source evidence for review.	◷ Calibration, usefulness, and clinician trust in confidence / provenance cues.
Clinician review workflow	✓ Enabled human review and reconciliation of AI-assembled content.	◷ Usability, clinician agreement, and accept / edit / reject patterns.
Standards-based output	✓ Produced clinician-reviewable PACIO ToC FHIR-aligned bundles.	◷ Formal conformance and robustness across broader real-world datasets.
Efficiency / clinician burden	— Not quantitatively measured in the proof of concept.	◷ Review time, manual abstraction effort, and burden versus baseline workflows.
Real-world / clinical outcomes	— Not evaluated in a production or clinical deployment.	◷ Workflow integration, delays / errors, transition quality, and clinical outcomes.

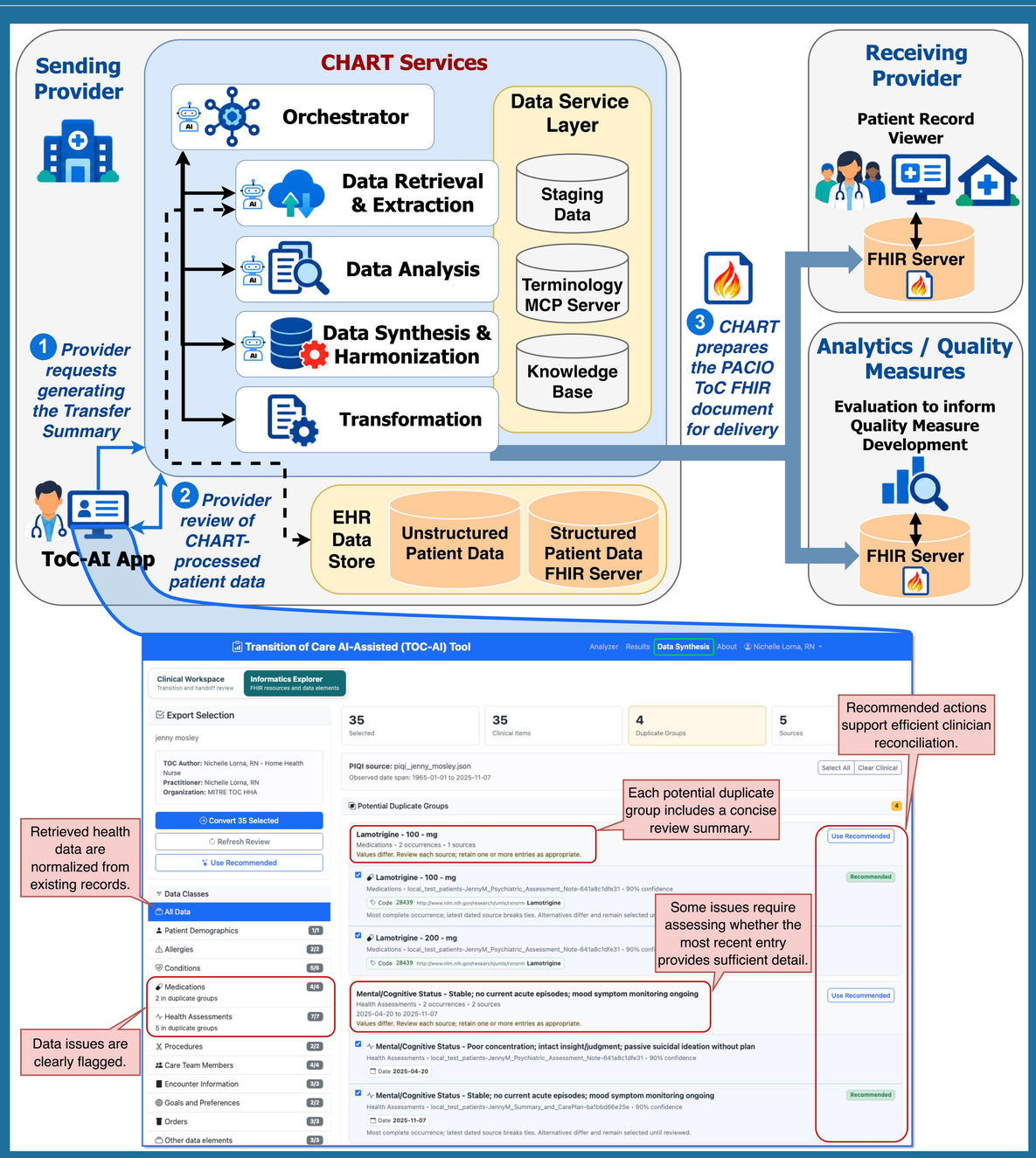
EVIDENCE BOUNDARY: **Synthetic-data feasibility demonstration only. No real-world deployment outcomes, efficiency gains, or clinical benefits were measured.**

Future exploration

Future research should evaluate extraction accuracy and completeness against PACIO ToC FHIR IG content expectations, conflict-detection precision, clinician agreement with AI suggestions, FHIR conformance, data quality, clinician review time, usability, and workflow integration.

CHART is a standards-led, AI-augmented toolkit of services that consolidate structured & unstructured data into standardized & harmonized healthcare information.

The **ToC-AI** App leverages CHART services to organize transition-relevant information while preserving clinician review, reconciliation, and source traceability, with the goal of producing complete, meaningful, and timely information for the next care team.



Clinician reconciliation workflow: AI-identified data issues, source evidence, and recommendations support human review before transition information is finalized for sharing.

Conclusions

This proof of concept demonstrates the feasibility of applying Generative AI to structured and unstructured clinical data to produce reviewable, transition-relevant information. PACIO and FHIR provide the interoperability foundation, while AI enhances the usability of standards-based exchange through semantic structuring and relevance filtering. **Human oversight remains essential for clinical review, reconciliation, safety, and trust.**

These conclusions are limited to technical feasibility and do not establish improvements in clinical outcomes, efficiency, or real-world workflow performance. Next steps include security and governance review, real-world validation, clinician usability studies, controlled pilot deployment, and benchmarking against baseline workflows. **The goal is not simply to exchange data, but to make interoperable data clinically meaningful, actionable, and ready for the next care team.**

Standards & Quality Approach

Standards and quality controls were applied throughout the ToC-AI workflow. The Analysis Agent used USCDI v5 and Terminology MCP services to identify and normalize clinical data using SNOMED CT, LOINC, RxNorm, and ICD-10-CM. The Data Synthesis Agent reconciled conflicting information and applied confidence scoring, while Patient Information Quality Improvement (PIQI) informed validation and scorecards enabled data-quality assessment. The Data Conversion Agent, supported by the PIQI-to-FHIR MCP service, transformed reconciled content into PACIO ToC-conformant FHIR bundles. A separate Quality Evaluation Agent assessed extraction quality, PIQI scorecards and ToC FHIR conformance.

Acknowledgements

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Contacts & Links

Collaborate with PACIO to evaluate and pilot AI-assisted, clinician-governed ToC workflows in real-world post-acute care settings.

Contact email:
info@pacioproject.org
www.PACIOProject.org

Read More About it!

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